

Q-Orbit — Phase 2 Manuscript Change Log

Document ID: QO-CHANGELOG-P2-001 | **Version:** 1.0 | **Date:** 2026-08-27 **Scope:** V1.0-RC1 (/mnt/agents/temp/Q-Orbit_Scientific_Research_Paper_Current.md, QO-SUB-RP-001) → **V1.0-RC2** (Q-Orbit_Scientific_Manuscript_V1.0-RC2.md). Companion artifacts: Q-Orbit_Manuscript_Claim_Ledger.md (QO-LEDGER-001), Q-Orbit_Figure_Specification.md (QO-FIGSPEC-001).

1. Basis and authoritative inputs

Phase 1 closed as **CONDITIONAL PASS — READY FOR PHASE 2 WITH DISCLOSED LIMITATIONS**. RC2 was produced exclusively from the Phase 1 authoritative basis: Consolidated Package, CFR v1.1, Canonical Audit, D3–D6, Final Red-Team Report, Closure Record, and the Interim Submission Package (QO-INTERIM-PKG-001, located at /mnt/agents/output/Q-Orbit_INTERIM_SUBMISSION_PACKAGE.md; a duplicate rendering exists as Q-Orbit_Preliminary_Submission_2026-08-27.md — both carry the correct negative median/minimum and are consistent with RC2). No new research, no new literature search, no canonical-number changes were performed or permitted.

2. Structural changes (RC1 → RC2)

- RC1’s 7-section layout expanded to the mandated 20-section submission structure (Title Block; Abstract; Keywords; Introduction; Research Question; Contribution Statement; Related Work; System and Protocol Model; Mathematical Framework; Methods; Results; Proof-to-Device Analysis; Proof-Profile Architecture; Security and Characterization Budget; Discussion; Limitations; Conclusion; Data and Reproducibility; References; Appendix — Claim Ledger).
- Manuscript length: ~13,100 content words; 77 verified references; 6 figure callouts with evidence-class captions (production per QO-FIGSPEC-001).

3. Corrections applied from the Phase 1 record

Change	Source
Grid median/minimum carried strictly NEGATIVE (−2,624.946810258186 / −3,828.414517626367 bits); RC1’ s sign-flipped table rendering explicitly quarantined in a §11.3 correction notice	CFR C-01…C-03
All canonical numbers set bit-exactly from CFR v1.1 (margin 41,338.62418456675; key 41,338; QBER_X 0.017422686665352745; _X 0.09270161340569935; 568/1,113/1,681; 0.33789411064842356; 12/12; 20/20; etc.)	CFR §2
Margin equation printed with the corrected 21-decomposition (+1 chain-rule constants, exact cancellation)	Red-team C2, D6 §4.2
_total = _c + _s + _char (+ _auth) with _char Σ_j _j defined once	Red-team C3
Upper-envelope disclosure on the screen partition/fraction; f_EC = 1 ideal-EC upper-bound disclosure	Red-team C8/C15, CFR N-07/N-08/N-26/N-27

Table 1 – continued

Change	Source
arXiv:2601.18035 cited as PREPRINT throughout; Quantum 10, 2037 appears only as the Wiesemann et al. record; Trényi & Curty retained solely as a documented misattribution (never cited for correlations)	Red-team C1/F-03; D3
Profile B framed conservatively (preprint anchor; imperfection integration is future work per its abstract; coverage rests on Optica Quantum 3, 525 (2025))	D4, red-team F-03
Claim-control: controlled labels on all major claims; §20 condensed claim ledger; REQ-01...05 disclosed as open limitations; prohibited-claims register honored	Closure Record §6; CFR §4/§6

4. Phase 2 QA gate (all checks executed before delivery)

1. **Canonical-number consistency** — **PASS** (mechanical battery: all mandated values present bit-exact; independent verifier: zero deviating decimals; derived quantities recomputed correct).
2. **Sign check** — **PASS** (11/11 median/minimum occurrences explicitly negative; positive renderings absent outside the correction notice).
3. **Equation check** — **PASS** (margin identity recomputed from printed components: diff 0.0; 21-decomposition algebraically exact; -accounting single-count).
4. **Citation hygiene** — **PASS after 2 MINOR fixes** (V-4.1: missing “preprint” label on one Kato citation in Table 7 — FIXED; V-4.2: undifferentiated “Tupkary et al. (2025)” key mapping to two records — FIXED with 2025a = arXiv:2502.10340 review / 2025b = Quantum 9, 1937, applied at all five in-text sites and both reference entries).
5. **Claim-control sweep** — **PASS after 2 fixes** (V-5.1 MAJOR: claim ledger’s manuscript-section pointers carried superseded rendering-B numbering — FIXED, all 56 affected rows re-mapped to true RC2 sections and the column key corrected; V-5.2 MAJOR: Figure 1 caption described a verification workflow while the figure specification mandates the per-pass finite-key computation pipeline — FIXED, caption replaced with the figspec’s exact text and the §10.1 callout aligned; V-5.3 COSMETIC: non-sanctioned caption token “documentation” — FIXED to the closed evidence-class vocabulary).
6. **Prohibited-claims sweep** — **PASS** (verifier: zero positive assertions of any register item; all occurrences are explicit negations/requirements; no characterization requirement phrased as a performed measurement).

Verifier verdict: FIXABLE findings, no BLOCKING defects; all five findings fixed and re-verified. Mechanical battery final state: 47/47 substantive PASS (one heuristic flag on two “certified”-in-requirement-context lines independently cleared by the verifier’s Check 6).

5. Judgment calls recorded by the writing stage

1. References verified in Phase 1 without recorded titles were cited author/venue/volume/year without inventing titles (fail-closed).

2. Tupkary 2026 preprints disambiguated as 2026a (arXiv:2601.18035) / 2026b (arXiv:2601.17960); Tupkary 2025 as 2025a/2025b (§4 above).
3. Reference set scoped to the 77 load-bearing D3-verified records; least-load-bearing citations removed under the word budget — all brief-mandated works retained.
4. Claim ledger runs to 65 rows (15 D5-effect rows + 10 BLOCKED rows mandated individually).
5. OPERATIONALLY-UNSUPPORTED used once (CL-55) for prohibited operational readings of the screen fraction, distinct from the BLOCKED register.

6. Standing disclosures carried into RC2

REQ-01...05 OPEN; no end-to-end re-execution; `_s/ _c` individually EV-9; artifact 01 EV-4/EV-1c; margins are upper bounds w.r.t. `f_EC`; screen partition is an upper-envelope deterministic quantity, not a probability; Profile-B anchor is a preprint; manuscript remains THEORETICAL / NOT PHYSICALLY VALIDATED.

7. Stop condition

D8 reviewer simulation NOT started, per instruction. Work stops at manuscript + supporting files. Website and prototype remain unbuilt.